College of Technology Galle

Course: Full Stack Software Development — Object Oriented Programming — Module K72T004M03 — NVQ Level 05

SEMESTER MODEL PAPER 08

Focus: Library/school style systems, interfaces, exam-style design

Difficulty: Moderate-Hard

Duration: 3 Hours **Total Marks:** 100 **Questions:** 7 structured questions (answer any **5**) **Each question:** 20 marks

Instructions to Candidates

1. Answer any **five (5)** questions out of the seven provided.
2. Each question carries **20 marks**. Maximum mark for this paper is **100**.
3. Write answers in the answer booklet. Do **not** write answers on this question paper.
4. Where Java code is required, write clear, syntactically correct Java as taught in this module (VS Code / JDK).
5. Show workings / short explanations where asked. Manage your time (about 35 minutes per question).
6. Mobile phones and programmable devices are not allowed.

Question 1 [20 Marks]

Full encapsulation review.

a. Critique a class that exposes public arrays of marks - what can go wrong? **(6 marks)**

b. Redesign with private data, controlled addMark/getAverage, and validation 0-100. **(10 marks)**

c. Write a short note on information hiding for a non-technical manager. **(4 marks)**

Question 2 [20 Marks]

Multi-level inheritance care.

a. Write inheritance chain Vehicle -> Car -> ElectricCar with one new field/method at each level. **(10 marks)**

b. Show constructor chaining with `super` at each level. **(6 marks)**

c. State one risk of deep inheritance trees and one alternative (e.g. composition). **(4 marks)**

Question 3 [20 Marks]

Interface segregation lite.

a. Define interfaces `Borrowable` and `Reservable` with suitable methods. **(6 marks)**

b. Implement a `Book` that is borrowable and reservable, and a `Magazine` that is only borrowable. **(10 marks)**

c. Why might two small interfaces be better than one large "LibraryItemOps" interface? **(4 marks)**

Question 4 [20 Marks]

Polymorphic collections.

a. Create an abstract `Staff` with `work()`; concrete `Librarian` and `Clerk`. **(8 marks)**

b. Store both in a `List<Staff>` (or array) and invoke `work()` in a loop. **(6 marks)**

c. Explain what would break if `work` were removed from the superclass and only defined in subclasses while the list type stayed `Staff`. **(6 marks)**

Question 5 [20 Marks]

Classroom composition.

a. Write `Desk` and a `Classroom` that owns a fixed array/list of desks created in its constructor. **(10 marks)**

b. Add `occupyDesk(int index, String studentName)` with bounds checks. **(6 marks)**

c. Why is this composition rather than a weak association? **(4 marks)**

Question 6 [20 Marks]

```
class A {
    void hello() { System.out.println("A"); }
}
class B extends A {
    void hello() { System.out.println("B"); }
    void onlyB() { System.out.println("onlyB"); }
}
```

Predict and reason.

a. For `A x = new B();` which calls compile: `x.hello()`, `x.onlyB()`? What prints for the legal call? **(8 marks)**

b. How can you call `onlyB` safely using a cast? What risk does a wrong cast bring? **(6 marks)**

c. Rewrite using an override-only design so clients need no cast for normal use. **(6 marks)**

Question 7 [20 Marks]

Exam scenario - COT library desk.

a. A librarian needs to search by title keyword and borrow by ISBN. Outline classes/methods (no full code). **(8 marks)**

b. Write the `matchesSearch(String keyword)` method body using `toLowerCase` / `contains`. **(6 marks)**

c. Explain where encapsulation and exceptions would improve this desk system. **(6 marks)**

******* END OF QUESTION PAPER *******